



Technische Universität München

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Master Thesis Exposé

A GraphQL API for the BIM Collaboration Format (BCF): Design, Implementation, and Evaluation

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1. General Problem Description



The Building Information Modeling (BIM) Collaboration Format (BCF) is a key standard maintained by buildingSMART International for issue tracking and information exchange in BIM coordination workflows. It enables stakeholders to communicate design or construction issues consistently across different software tools through standardized entities such as topics, comments, and viewpoints.

Currently, BCF relies primarily on a RESTful API architecture. While functional, REST exhibits notable limitations in complex, data-intensive workflows. These include over-fetching, under-fetching, and endpoint proliferation, which hinder performance and developer efficiency when handling nested BIM data structures.

Modern software applications increasingly rely on flexible data exchange mechanisms. GraphQL, a data query language developed by Meta, offers a promising alternative by allowing clients to specify precisely the data structure they need through a single endpoint. Its introduction into BCF systems could improve communication efficiency, API usability, and interoperability between BIM tools.

This thesis thus aims to investigate whether GraphQL can provide a technically viable, performant, and scalable alternative to REST in the context of the BCF specification. The work combines academic exploration with industrial validation through collaboration with ALLPLAN GmbH, developer of the BIMplus platform.

2. Summary of Related Work

The evolution of API technologies and their application in the AECO industry has generated substantial research examining efficient data exchange mechanisms for BIM. This section reviews literature addressing GraphQL as an alternative to REST, the BCF, and their integration potential.

- GraphQL as an API Technology

GraphQL has emerged as a compelling alternative to traditional REST architectures, addressing core limitations in data exchange flexibility and efficiency. Brito and Valente (2020) conducted a controlled experiment comparing REST and GraphQL APIs, demonstrating that GraphQL reduces data over-fetching by enabling clients to specify exact data requirements, resulting in smaller payload sizes and reduced network overhead. Their findings confirm GraphQL's advantage in scenarios involving complex, nested data structures characteristics prevalent in BIM workflows.



Lawi, Panggabean, and Yoshida (2021) extended performance analysis to massive, intensive access environments, evaluating GraphQL's scalability under high query loads. While GraphQL demonstrated flexibility advantages, the study highlighted computational trade-offs in resolver execution, particularly for deeply nested queries. Vadlamani et al. (2021) further investigated whether GraphQL could fully replace REST, concluding that while GraphQL excels in reducing redundant data transfer and improving developer experience, REST remains advantageous for simpler, cacheable queries with predictable access patterns. However, it should be noted that both server-side and client-side caching are achievable in GraphQL implementations, though they require more sophisticated configuration compared to standard HTTP caching mechanisms used in REST APIs (Apollo GraphQL, 2024).

- BCF and BIM Collaboration

The BIM Collaboration Format, standardized by buildingSMART International, facilitates issue tracking and model-based communication across heterogeneous BIM tools. Schulz and Beetz (2021) explored BCF's application in server-based workflows for existing building documentation, identifying critical limitations in the RESTful BCF API. Their work revealed that BCF's nested hierarchical structure hinders efficient information retrieval, requiring multiple sequential requests to access related resources such as topics, viewpoints, and comments. This N+1 query problem directly motivates investigation of alternative API paradigms like GraphQL.

Schulz, Oraskari, and Beetz (2021, 2023) developed bcfOWL, an ontology representing BCF data using RDF and OWL to enable semantic queries via SPARQL. While this approach demonstrated the feasibility of graph-based BCF representation and enhanced integration with Linked Data ecosystems, their subsequent reflections (2023) acknowledged practical challenges including SPARQL query complexity and performance limitations in production environments. These findings suggest that alternative graph-based technologies like GraphQL may offer more accessible solutions for practical BCF workflows while retaining structural advantages.

- GraphQL in BIM Contexts

Werbrouck et al. (2019) investigated semantic query languages for knowledge-based web services in construction, proposing context-expanded GraphQL queries to access heterogeneous Linked Building Data. Their approach demonstrated GraphQL's capability to query distributed BIM resources through federated schemas, though implementation remained experimental.



IFC2GRAPHQL: Schema mapping and automated API construction from IFC to GraphQL (Wolf, Esser & Borrmann 2025) introduced a framework that automatically maps the IFC schema (EXPRESS) to a GraphQL schema and generates a corresponding API. Their prototype shows that GraphQL's typed schema and nested-query capabilities align well with IFC's object-oriented/hierarchical structure, enabling finer-grained access to building elements rather than full model downloads. While the authors present feasibility rather than full deployment, this work lays groundwork for applying GraphQL in BIM data exchange contexts.

- Semantic Web Alternatives

Beyond GraphQL, the Linked Data community has explored SPARQL-based approaches for BIM interoperability. These methods offer powerful semantic reasoning capabilities but impose steeper learning curves and require RDF triple store infrastructure. The bcfOWL work by Schulz et al. (2023) serves as a critical reference point, demonstrating both the potential and practical limitations of Semantic Web approaches for BIM collaboration workflows.

3. Research Gap and Preliminary Hypotheses

The identified gap lies in the lack of technical exploration of GraphQL for BCF-based issue tracking. The thesis will investigate whether GraphQL can increase performance and interoperability compared to REST APIs currently implemented in BCF workflows.

Hypotheses:

1. The BCF data model can be represented effectively using GraphQL type definitions without loss of semantic fidelity.
2. A GraphQL-based API will demonstrate improved efficiency in data exchange, reducing payload size and query overhead compared to REST implementations.
3. GraphQL simplifies developer interaction and maintenance by consolidating multiple endpoints into a single, schema-driven query structure.
4. Industrial testing within the ALLPLAN environment can validate its integration feasibility for future BCF-based interoperability standards.

4. Intended Methodology



- Schema Design
 - Model BCF entities (topics, comments, viewpoints) as GraphQL types based on BCF version 2.1 for start
 - Design a comprehensive GraphQL schema that represents the complete BCF data model with appropriate queries, mutations, and subscriptions
 - Document schema design decisions and their alignment with BCF specification requirements
- Standalone Server Implementation
 - Develop an independent GraphQL server implementation
 - Implement data persistence layer to store BCF data
 - Create resolver logic for all GraphQL operations (queries and mutations) that directly manage BCF entities
- REST API Implementation (Comparison Baseline)
 - Implement a parallel RESTful API following the official BCF API specification for the same version
 - Use identical database and business logic layer to ensure fair comparison
- IFC Integration Exploration
 - Investigate potential integration points between BCF topics/viewpoints and IFC model elements
 - Explore GraphQL schema extensions to query related IFC geometry or properties referenced in BCF viewpoints
 - If feasible, implement a proof-of-concept connection demonstrating how GraphQL could bridge BCF issue tracking with IFC model data
- Testing and Dataset Preparation
 - Prepare representative BCF datasets covering various complexity levels (simple topics, nested comments, multiple viewpoints)
 - Develop automated test suites for both GraphQL and REST implementations



- Performance Evaluation
 - Response time and latency
 - Payload size and network efficiency
 - Query complexity handling (nested vs. simple queries)
 - Concurrent request handling and scalability
- Validation with ALLPLAN
 - Present prototype and evaluation results during expert meetings
 - Gather industry feedback on practical applicability and integration feasibility
 - Discuss potential adoption barriers and implementation considerations for production BIM platforms

5. Industrial Collaboration with ALLPLAN

The thesis will be carried out in cooperation with ALLPLAN GmbH, a recognized provider of BIM software solutions. The company supports the research by:

- Providing API access to the BIMplus staging environment and relevant documentation.
- Offering mentorship (approximately four to five expert meetings) to define direction.

This collaboration ensures that the research outcomes are tested against authentic BIM system conditions and that the results are directly relevant to industrial development.

6. Expected Outcomes

The project will deliver a functioning GraphQL implementation of the BCF API, benchmarked against the RESTful specification on which BIMplus currently relies. The outcomes will include:

- A formal schema model mapping BCF to GraphQL types.
- A prototype middleware demonstrating how GraphQL can interface with existing BIM APIs.
- A comparative performance assessment confirming technical and practical viability.



The resulting insights will inform both academic understanding and industry practices, contributing to the modernization of BIM collaboration standards and potential adoption within BIM platforms, standardization initiatives and commercial environments.

7. Preliminary Timetable

Phase	Duration	Description
Nov 20 – Dec 5	2 weeks	Administrative setup and dataset preparation
Dec 6, 2025 – Jan 16	6 weeks	Schema design and GraphQL server implementation
Jan 17 – Feb 13,	4 weeks	REST API implementation
Feb 14 – Mar 13	4 weeks	Testing framework and performance evaluation
Mar 14 – Apr 17	5 weeks	Results analysis and industry validation
Apr 18 – May 20	5 weeks	Documentation and thesis finalization

8. Resources

BCF REST API specification: <https://github.com/buildingSMART/BCF-API>

BuildingSmart foundation API

<https://github.com/buildingSMART/foundation-API>

BCF XML schema: <https://github.com/buildingSMART/BCF-XML>

BcfServer <https://github.com/Design-ComputationRWTH/bcfServer>

GraphQL Specification: <https://spec.graphql.org/October2021>



ALLPLAN BCF

<https://doc.allplan.com/spaces/bimpluspublic/pages/18040737/4.11+BCF+Service>

9. References

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